From Individual EVs to Physical Chargers: Physics-Constrained Flexibility Forecasting and Model Predictive Control

Matched-scope prediction, exact aggregation, and continuous demand-charge accounting for workplace charging

Yizhan Gu
University of California San Diego, La Jolla, CA, USA

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Abstract

This paper asks whether the physical charger is a more useful prediction and control entity than the individual electric vehicle (EV) for unidirectional workplace charging. The setting excludes vehicle-to-grid export, battery storage, photovoltaics, and demand response; economic value therefore means avoided charging cost rather than market revenue. We construct two genuinely different next-day forecasting tasks and prevent a common comparison error by evaluating individual-EV and charger models on exactly the same sessions. A second operational metric retains all charger demand and penalizes EV demand that cannot be assigned to a driver observed often enough during training. In a chronological July–September 2023 test, the recurring-driver cohort represents 29.77% of full energy. On this identical matched scope, charger aggregation reduces aggregate daily energy mean absolute error by 57.95% for DLinear, 70.94% for LSTM, 73.02% for TCN, and 69.16% for iTransformer, averaged over three seeds. A seasonal-naive forecast is identical at both resolutions, providing an algebraic and numerical scope check. We further introduce a set-free flexibility signature comprising cumulative deliverable energy bounds and physical-port occupancy at six intraday anchors. A differentiable output layer guarantees monotonicity, ordered bounds, terminal energy equality, and occupancy-limited capacity. On the physically screened matched cohort, the physics-iTransformer charger model reduces terminal-energy, lower-envelope, and upper-envelope MAE by 69.19%, 60.71%, and 64.07% relative to the same EV architecture; all paired block-bootstrap intervals exclude zero. We then derive a sparse charger-level linear program whose cumulative-energy envelopes are an exact projection of the session-level problem when sessions do not overlap on a physical port. A separate forecast-calibrated MPC pilot carries executed demand peaks continuously through each billing month. Across the same 92 test days, conformal MPC costs $4,177.90, 28.23% below immediate charging and 1.47% below point-forecast MPC, with zero unserved energy. The new anchor signatures have not yet been lifted to full-resolution uncertainty sets and coupled to MPC; consequently, their economic value remains a pre-registered next experiment rather than a result of this draft.

Keywords: electric-vehicle charging; charger-based forecasting; individual-EV forecasting; graph neural network; iTransformer; model predictive control; demand charges

1 Introduction

Workplace EV charging is naturally a receding-horizon control problem: sessions arrive throughout the day, their flexibility differs, and monthly demand charges couple otherwise independent days. Session-level optimization is physically transparent but expands with the number of vehicles. Charger-level optimization is smaller, but a daily-energy-only aggregation can move energy across a vehicle's arrival or deadline and consequently report false savings. The first question in this work is therefore structural: under which operating conditions is a charger representation both smaller and exactly feasible?

The second question concerns prediction resolution and uncertainty. Classical charging formulations optimize known requests [3, 12], whereas closed-loop station control must act before later sessions are observed. Stochastic EV-station models explicitly consider uncertain arrival, departure, and energy [8]. Recent charging-hub MPC combines machine-learning forecasts with conformal uncertainty sets and reports a roughly 1% improvement over deterministic point forecasts in a richer PV–battery setting [2]. Here we isolate the charger-control effect: there is no storage, PV, export, V2G, or demand-response signal. Importantly, changing an optimization index after making one shared forecast does not compare EV and charger forecasting. Each representation must define its own prediction entities, training panel, and error process.

Our contributions are:

1. a leakage-safe, matched-scope protocol that compares individual-EV and charger forecasts on exactly the same realized sessions and separately quantifies open-set driver coverage;
2. a common model ladder spanning seasonal naive, ridge, hurdle regression, DLinear, LSTM, TCN, iTransformer, and a charger graph-temporal regressor;
3. a session-matching-free flexibility target and differentiable physical output layer that predicts cumulative energy bounds and port occupancy while satisfying feasibility by construction;
4. an exact sparse charger projection for nonoverlapping physical-port sessions, together with an explicit disaggregation argument;
5. a rolling MPC implementation that carries executed billing-peak states across days and evaluates only incremental monthly demand cost;
6. a leakage-controlled control comparison of no forecast, persistence, historical-median, one-sided conformal, and perfect-information forecasts; and
7. numerical invariance, feasibility, predictive coverage, paired block-bootstrap, and scalability audits that make the boundary of the current evidence explicit.

2 Related work and research gap

2.1 Time-series forecasting architectures

Strong modern forecasting comparisons must include simple models. DLinear decomposes a series into moving-average trend and seasonal residuals before linear projection and has challenged the assumption that deeper Transformers are automatically superior [15]. PatchTST uses channel-independent patch tokens to extend the effective history at lower attention cost [7]. iTransformer instead embeds the complete history of each physical variable as a token and applies attention across variables [5]. This inversion is natural for the five correlated session attributes used here. SAMformer further shows that shallow attention, reversible normalization, and sharpness-aware optimization can be competitive in small-data regimes [4]. We therefore treat architecture complexity as an experimental factor, not as the scientific contribution.

2.2 Spatio-temporal charging-demand forecasting

Hybrid ML has previously been used to forecast anonymous workplace-EV arrival, plug-duration, and energy attributes for online MPC [6]. That work predicts a variable-cardinality individual-session set. We do not reproduce its session assignment metric or historical-session retrieval procedure; our question is different: whether the physical charger is itself a statistically and operationally preferable prediction entity. Accordingly, our primary targets are additive daily quantities and cumulative feasible-energy sets that can be compared without pairing forecasted and realized EVs.

Graph WaveNet learns latent adjacency jointly with dilated temporal convolutions [14], motivating graph models when charging ports share location, calendar, or mobility structure. Recent EV demand preprints combine station graphs with multi-frequency or large-model features [1] and graph convolutions with traffic, weather, and temporal encoders [13]. These studies reinforce the importance of station-level spatial structure, but do not isolate the statistical effect of changing the prediction entity from individual EV to physical charger on an identical set of sessions. Our matched-scope design targets that gap. It also exposes a deployment issue that aggregate station-demand studies can hide: a driver-specific forecaster cannot identify a driver absent from its training history without reservation, fleet, or mobility information.

3 Forecasting at two entity resolutions

3.1 Daily targets and chronological information

Let n denote either a driver in the EV task or a physical station–port pair in the charger task. The next-day target is

$$\mathbf{y}_{t,n} = [c_{t,n}, e_{t,n}, a_{t,n}, d_{t,n}, \ell_{t,n}]^\top, \tag{1}$$

where c is session count, e is total energy, a and d are mean arrival and departure slots, and ℓ is mean dwell duration. Timing features are undefined and masked when $c_{t,n} = 0$. Both tasks use the same feature semantics, 28-day lookback, and calendar sine/cosine covariates.

All selection and normalization end with the training interval. The driver cohort for minimum history h is

$$\mathcal{D}_h = \left\{ i : \sum_{t \in \mathcal{T}_{\text{train}}} c_{t,i} \geq h \right\}. \tag{2}$$

The main forecast experiment fixes $h = 3$, yielding 372 drivers. The charger set is the same six physical ports used by the MPC pilot. The data windows are January 2022–March 2023 for training, April–June 2023 for validation, and July–September 2023 for testing. The test period does not determine driver eligibility, scaling, graph edges, early stopping, or any hyperparameter.

3.2 Matched and operational estimands

Comparing per-entity errors would be invalid because a charger aggregates more energy than a driver. We instead define a matched charger panel using only sessions whose drivers belong to $\mathcal{D}_h$. Hence, for every day,

$$\sum_{i \in \mathcal{D}_h} e_{t,i} = \sum_{c \in \mathcal{C}} e^{\text{match}}_{t,c}. \tag{3}$$

Both models are evaluated after summing their predictions to daily energy. The matched daily loss is

$$L^r_{t,m} = \left| \sum_{n \in \mathcal{N}_r} e_{t,n} - \sum_{n \in \mathcal{N}_r} \widehat{e}^m_{t,n} \right|, \quad r \in \{\text{EV}, \text{charger}\}. \tag{4}$$

Equation (3) guarantees a common target and denominator.

Matched scope does not describe deployment coverage. Let full six-port energy be E^{full}_t, matched energy be E^{match}_t, and unmodelled energy be $U_t = E^{\text{full}}_t - E^{\text{match}}_t \geq 0$. An overprediction for one known driver cannot substitute for an unseen driver on another physical session. We therefore define the no-substitution operational EV loss

$$L^{\text{EV,op}}_{t,m} = \left| E^{\text{match}}_t - \widehat{E}^{\text{EV},m}_t \right| + U_t, \tag{5}$$

and operational WAPE

$$\text{WAPE}^{\text{EV,op}}_m = \frac{\sum_t L^{\text{EV,op}}_{t,m}}{\sum_t E^{\text{full}}_t}. \tag{6}$$

This prevents aggregate cancellation between fictitious energy assigned to an absent known driver and energy belonging to an unidentified driver.

3.3 Models and objectives

The common ladder comprises seasonal naive, ridge regression, a two-stage hurdle model, DLinear, LSTM, TCN, and iTransformer. The seasonal-naive prediction is $\widehat{\mathbf{y}}_{t,n} = \mathbf{y}_{t-7,n}$. Because summation and the seven-day lag commute, its EV and charger aggregate predictions must be identical under (3); this is an explicit pipeline check.

Let μ_f^{tr} and σ_f^{tr} denote feature statistics computed on training entity-days only. The causal input tensor is

$$\widetilde{X}_{t,n,\ell,f} = \frac{X_{t,n,\ell,f} - \mu_f^{\mathrm{tr}}}{\sigma_f^{\mathrm{tr}}}, \qquad \ell = 1, \ldots, L, \tag{7}$$

with $L = 28$. The next-day calendar vector contains sine/cosine encodings of weekday and day of year. Neural models minimize

$$\mathcal{L}(\theta) = \frac{\sum_{t,n,f} m_{t,n,f} (\widehat{y}_{t,n,f} - y_{t,n,f})^2}{\sum_{t,n,f} m_{t,n,f}}, \tag{8}$$

where count and energy receive weight three on active days and timing targets are masked on inactive days. Scaling statistics use training data only.

For iTransformer, the complete lookback of variable f becomes one token:

$$\mathbf{z}_f^{(0)} = W_x [\widetilde{x}_{t-L+1,f}, \ldots, \widetilde{x}_{t,f}]^\top + \mathbf{e}_f, \tag{9}$$

$$Q_h = Z^{(0)} W_h^Q, \quad K_h = Z^{(0)} W_h^K, \quad V_h = Z^{(0)} W_h^V, \tag{10}$$

$$A_h = \mathrm{softmax}\left(\frac{Q_h K_h^\top}{\sqrt{d_h}} \right) V_h. \tag{11}$$

The four head outputs are concatenated, passed through residual normalization and a feature-wise feed-forward network, and denoted by $\mathbf{H} \in \mathbb{R}^{F \times 32}$. With one encoder layer and feed-forward width 128, the unconstrained next-day logits are

$$\mathbf{z} = W_o \mathbf{H} + W_q \mathbf{q}_t, \qquad \mathbf{z} \in \mathbb{R}^F. \tag{12}$$

Thus attention operates across physical attributes rather than timestamps, while the 28-point temporal trajectory is retained inside each variate token.

The charger-only graph model first encodes each port history with a shared GRU. Its adjacency uses the top three absolute energy correlations computed only on training days. After adding self-loops and symmetric normalization, $\widetilde{A} = D^{-1/2}(A + I)D^{-1/2}$, one graph layer gives

$$\mathbf{G}_t = \sigma(W_s \mathbf{H}_t + W_g \widetilde{A} \mathbf{H}_t + W_q \mathbf{q}_t), \qquad \widehat{\mathbf{Y}}_{t+1} = W_o \mathbf{G}_t, \tag{13}$$

where $\mathbf{q}_t$ contains calendar covariates.

For stochastic models we report three deterministic seeds. Daily paired differences $L_{t,m}^{\mathrm{EV}} - L_{t,m}^{\mathrm{charger}}$ are summarized after averaging seed-specific losses by day. A moving-block bootstrap with seven-day blocks and 5,000 resamples provides 95% intervals; these intervals address short serial dependence but do not replace replication across sites or seasons.

3.4 Set-free flexibility-envelope target

Daily means in (1) do not fully describe when predicted energy can be delivered. We therefore introduce an additive feasibility signature at anchor slots

$$\mathcal{B} = \{32, 48, 64, 80, 84, 96\},$$

corresponding to 08:00, 12:00, 16:00, 20:00, 21:00, and 24:00. Let τ_b be anchor b, $\tau_0 = 0$, and $\mathcal{K}_b = \{\tau_{b-1}+1, \ldots, \tau_b\}$. For session i, the maximum energy deliverable by anchor b and minimum energy required by that anchor are

$$U_{i,b} = \min\big\{E_i, \bar{p}_i \Delta t \,[\min(\tau_b, d_i) - a_i + 1]_+\big\}\,, \tag{14}$$

$$L_{i,b} = \max\big\{0, E_i - \bar{p}_i \Delta t \,[d_i - \max(\tau_b + 1, a_i) + 1]_+\big\}\,. \tag{15}$$

The occupied-port equivalent in block b is

$$O_{i,b} = \frac{1}{|\mathcal{K}_b|} \sum_{t \in \mathcal{K}_b} \mathbf{1}\{a_i \le t \le d_i\}. \tag{16}$$

For entity n, the target $\mathbf{s}_{t,n} = [\mathbf{L}_{t,n}, \mathbf{U}_{t,n}, \mathbf{O}_{t,n}]$ is obtained by summing (14)–(16) over its sessions. Because every component is session-additive, the same-session identity extends beyond daily energy:

$$\sum_{i \in \mathcal{D}_h} \mathbf{s}_{t,i} = \sum_{c \in \mathcal{C}} \mathbf{s}_{t,c}^{\text{match}}. \tag{17}$$

No forecast-to-realized session assignment is needed.

An unconstrained regressor can output decreasing cumulative energy, reversed bounds, or terminal energy exceeding predicted occupancy capacity. Partition its 18 logits as $\mathbf{z} = [\mathbf{z}^L, \mathbf{z}^U, \mathbf{z}^O]$. The physics-iTransformer maps occupancy through $\widehat{O}_b = \text{sigmoid}(z_b^O)$ and defines block and cumulative capacities

$$\widehat{K}_b = \bar{p}\Delta t |\mathcal{K}_b| \widehat{O}_b, \qquad \widehat{C}_b = \sum_{j \le b} \widehat{K}_j, \qquad \widehat{R}_b = \widehat{C}_B - \widehat{C}_b. \tag{18}$$

The terminal-energy logit reuses both terminal curve channels, $z^E = (z_B^L + z_B^U)/2$, and therefore

$$\widehat{E} = \text{sigmoid}(z^E)\widehat{C}_B. \tag{19}$$

For $r \in \{L, U\}$, softmax fractions and their cumulative targets are

$$\pi_b^r = \frac{\exp(z_b^r)}{\sum_{j=1}^{B} \exp(z_j^r)}, \qquad \widetilde{r}_b = \widehat{E} \sum_{j \le b} \pi_j^r, \qquad \widehat{M}_b = \Big[\widehat{E} - \widehat{R}_b\Big]_+. \tag{20}$$

The differentiable closure implemented by cumulative maxima is

$$\widehat{U}_b = \max_{k \le b} \min\{\widehat{C}_k, \max(\widetilde{U}_k, \widehat{M}_k)\}, \tag{21}$$

$$\widehat{L}_b = \max_{k \le b} \min\{\widehat{U}_k, \max(\widetilde{L}_k, \widehat{M}_k)\}, \tag{22}$$

with both terminal components set exactly to $\widehat{E}$. Consequently,

$$\begin{gathered} 0 \le \widehat{L}_1 \le \cdots \le \widehat{L}_B = \widehat{E}, \\ 0 \le \widehat{U}_1 \le \cdots \le \widehat{U}_B = \widehat{E}, \\ \widehat{L}_b \le \widehat{U}_b \le \widehat{C}_b, \qquad \widehat{L}_b \ge \widehat{E} - \widehat{R}_b. \end{gathered} \tag{23}$$

Thus every network output belongs to the compressed feasibility cone during training and inference. It is transformed back to standardized coordinates inside the network, and the signature objective is

$$\mathcal{L}_{\text{sig}}(\theta) = \frac{\sum_{t,n,f} w_{t,n} [(\widehat{s}_{t,n,f} - s_{t,n,f})/\sigma_f^{\text{tr}}]^2}{F \sum_{t,n} w_{t,n}}, \qquad w_{t,n} = \begin{cases} 3, & E_{t,n} > 0, \\ 1, & E_{t,n} = 0. \end{cases} \tag{24}$$

Zero days remain in the objective because absence is informative. Primary errors are computed after summing signatures across entities: terminal-energy MAE and mean absolute lower- and upper-curve error across anchors.

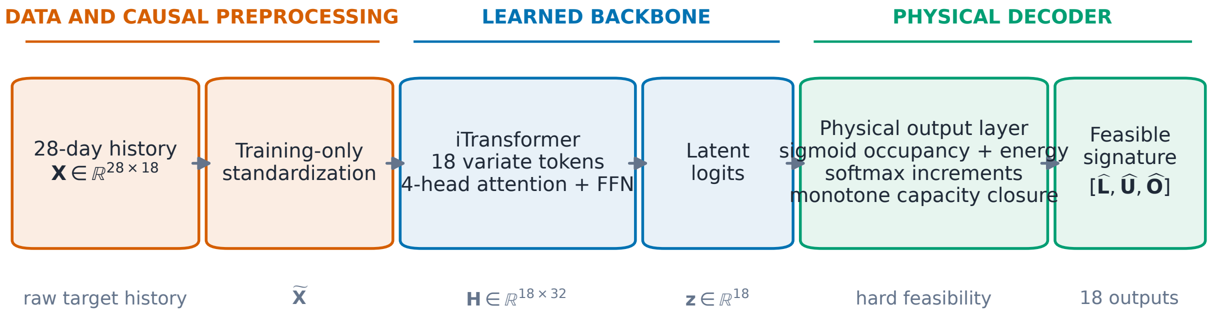

Figure 1: Physics-iTransformer data path. The learned backbone predicts 18 unconstrained logits; the differentiable decoder in (18)–(22) converts them to a feasible next-day flexibility signature during both training and inference.

Table 1: Physics-iTransformer specification. The same architecture, optimizer, history, and target semantics are used for individual-EV and charger tasks; only the prediction entity and resulting panel change.

Stage	Mathematical object	Implementation	Role
Target	$[\mathbf{L}, \mathbf{U}, \mathbf{O}] \in \mathbb{R}^{18}$	six anchors, three channels	additive feasible set
History	$\widetilde{\mathbf{X}} \in \mathbb{R}^{28\times18}$	training-only scaling	leakage-safe context
Backbone	18 variate tokens in $\mathbb{R}^{32}$	one 4-head encoder layer	cross-feature dependence
Physical head	$(\widehat{\mathbf{L}}, \widehat{\mathbf{U}}, \widehat{\mathbf{O}})$	sigmoid, softmax, cumulative closure	hard feasibility
Objective	weighted standardized MSE	active weight 3	retain informative zero days

4 Physical and tariff model

4.1 Sets, data, and assumptions

One day is divided into $\mathcal{T} = \{1, \ldots, T\}$ intervals of length $\Delta t = 0.25$ h; $T = 96$. Let $\mathcal{C}$ be the physical chargers and $\mathcal{I}_c$ the sessions assigned to charger c. Session i has arrival a_i, inclusive departure d_i, requested grid energy E_i, and maximum power $\bar{p}_i$. Its availability set is

$$\mathcal{T}_i = \{t \in \mathcal{T} : a_i \le t \le d_i\}.$$

Charging is unidirectional:

$$p_{i,t} \ge 0.$$

There is no export, stationary battery, PV, feeder-network constraint, or demand-response payment. Charging efficiency is already absorbed into the metered energy request. Once a session arrives, $d_i, E_i, \bar{p}_i$ are assumed observable. Most importantly, sessions on each selected physical port do not overlap:

$$i, j \in \mathcal{I}_c,\ a_i < a_j \quad \Longrightarrow \quad d_i < a_j. \tag{25}$$

Equation (25) is checked in data and before every optimization.

4.2 Session-level reference problem

The exact EV-level feasible set is

$$0 \le p_{i,t} \le \bar{p}_i \mathbf{1}_{\{t \in \mathcal{T}_i\}}, \qquad i \in \mathcal{I},\ t \in \mathcal{T}, \tag{26a}$$

$$\Delta t \sum_{t \in \mathcal{T}_i} p_{i,t} = E_i, \qquad i \in \mathcal{I}. \tag{26b}$$

The total site charging power is

$$P_t = \sum_{i \in \mathcal{I}} p_{i,t}. \tag{27}$$

This reference formulation is a linear program (LP) and is used only for invariance and scalability comparisons, not for the rolling paper experiment.

4.3 Exact charger-level cumulative envelopes

Define cumulative delivered energy through interval t for session i. The greatest energy that could have been delivered is

$$U_{i,t} = \min\left\{E_i,\ \bar{p}_i \Delta t\ [\min(t, d_i) - a_i + 1]_+\right\}, \tag{28}$$

and the least energy that must already have been delivered, leaving enough future capacity, is

$$L_{i,t} = \max\left\{0,\ E_i - \bar{p}_i \Delta t\ [d_i - \max(t+1, a_i) + 1]_+\right\}. \tag{29}$$

Aggregate envelopes are $U_{c,t} = \sum_{i \in \mathcal{I}_c} U_{i,t}$ and $L_{c,t} = \sum_{i \in \mathcal{I}_c} L_{i,t}$. With charger power $p_{c,t}$, introduce the sparse cumulative state

$$e_{c,0} = 0, \tag{30a}$$

$$e_{c,t} = e_{c,t-1} + \Delta t\, p_{c,t}, \qquad c \in \mathcal{C},\ t \in \mathcal{T}, \tag{30b}$$

$$L_{c,t} \le e_{c,t} \le U_{c,t}, \qquad c \in \mathcal{C},\ t \in \mathcal{T}, \tag{30c}$$

$$0 \le p_{c,t} \le \sum_{i \in \mathcal{I}_c} \bar{p}_i \mathbf{1}_{\{t \in \mathcal{T}_i\}}. \tag{30d}$$

Under (25), at most one term in the right side of (30d) is nonzero.

Proposition 1 (Exact charger projection)**.** *If sessions on every physical port satisfy* (25)*, the set of aggregate trajectories satisfying* (30) *is exactly the projection of the session-level feasible set* (26) *under* $p_{c,t} = \sum_{i \in \mathcal{I}_c} p_{i,t}$.

Proof. For any feasible session dispatch, summing its cumulative energy establishes (30b)–(30c); availability establishes (30d). Conversely, nonoverlap means every occupied charger–time pair has one unique active session. Assign $p_{c,t}$ to that session and zero to all others. The upper envelope prevents premature energy, the lower envelope forces each completed session to have received E_i, and the pointwise bound enforces $\bar{p}_i$. At $t = d_i$, both session envelopes equal E_i; because later sessions have not arrived, their cumulative contribution is zero. Hence each session energy equality holds and the assignment is a feasible disaggregation. □

The recurrence (30b) has $O(|\mathcal{C}|T)$ nonzeros. A dense prefix-sum encoding has $O(|\mathcal{C}|T^2)$ nonzeros even though it represents the same state.

4.4 Continuous monthly tariff state

Let λ_t be the time-of-use energy price and $\mathcal{T}^{\text{on}} = \{65, \ldots, 84\}$ the 16:00–21:00 on-peak intervals. The implemented summer tariff uses $\lambda_t =$ \$0.12628/kWh on peak and \$0.10679/kWh otherwise, noncoincident demand rate $\kappa^N =$ \$24.48/kW, and on-peak demand rate $\kappa^P =$ \$28.92/kW. Additional tariff coefficients are a fraction $\rho = 0.0578$ and volumetric charge $\eta =$ \$0.00668/kWh.

For billing month m, let $z^N_{m,d-1}$ and $z^P_{m,d-1}$ be the *executed* peaks before day d. Their recursion is

$$z^N_{m,d} = \max\left\{ z^N_{m,d-1}, \max_{t\in\mathcal{T}} P_{d,t} \right\}, \tag{31a}$$

$$z^P_{m,d} = \max\left\{ z^P_{m,d-1}, \max_{t\in\mathcal{T}^{\text{on}}} P_{d,t} \right\}. \tag{31b}$$

The day-d incremental bill contribution optimized and evaluated is

$$\begin{aligned} J_{m,d} = {} & (1+\rho)\Delta t \sum_{t\in\mathcal{T}} \lambda_t P_{d,t} \\ & + (1+\rho)\kappa^N (z^N_{m,d} - z^N_{m,d-1}) \\ & + (1+\rho)\kappa^P (z^P_{m,d} - z^P_{m,d-1}) \\ & + \eta\Delta t \sum_{t\in\mathcal{T}} P_{d,t}. \end{aligned} \tag{32}$$

Thus demand increments telescope:

$$\sum_{d\in m} \kappa^N (z^N_{m,d} - z^N_{m,d-1}) = \kappa^N z^N_{m,D_m},$$

and similarly on peak. Treating each day as a fresh bill would instead charge the demand maximum repeatedly and is not used in reported results.

5 Forecast-calibrated rolling MPC

5.1 Causal point forecast

For target day d, the historical-median model uses only the same weekday in the preceding $K = 4$ weeks. For each charger c, it predicts the number of sessions by the rounded median of the four counts. Sessions are ordered by arrival on each historical day. For ordinal j, the point forecast is

$$(\widehat{a}_{cjd}, \widehat{d}_{cjd}, \widehat{E}_{cjd}, \widehat{\bar{p}}_{cjd}) = \text{median}_{\ell=1,\ldots,K}(a_{cj,d-7\ell}, d_{cj,d-7\ell}, E_{cj,d-7\ell}, \bar{p}_{cj,d-7\ell}), \tag{33}$$

using the available ordinal samples. Windows are clipped to the day and sequentially reconciled to retain port nonoverlap.

5.2 One-sided split-conformal correction

We calibrate failure-relevant residuals on ordinal-matched sessions:

$$r^a_k = a_k - \widehat{a}_k, \qquad r^d_k = \widehat{d}_k - d_k, \qquad r^E_k = E_k - \widehat{E}_k. \tag{34}$$

Positive values respectively mean later arrival, earlier departure, or larger energy than forecast. For a residual sample $r_1, \ldots, r_n$, the finite-sample one-sided split-conformal quantile is

$$q_{1-\alpha}(r) = r_{(\,k\,)}, \qquad k = \min\{n, \lceil (n+1)(1-\alpha) \rceil\}, \tag{35}$$

where $r_{(k)}$ is the k-th order statistic [10, 11]. The robust session passed to the deterministic MPC is

$$\widetilde{a}_i = \min\{T, \widehat{a}_i + \max(0, q^a)\}, \tag{36a}$$

$$\widetilde{d}_i = \max\{\widetilde{a}_i, \widehat{d}_i - \max(0, q^d)\}, \tag{36b}$$

$$\widetilde{E}_i = \min\{\widehat{E}_i + \max(0, q^E), \bar{p}_i \Delta t(\widetilde{d}_i - \widetilde{a}_i + 1)\}. \tag{36c}$$

This is a conservative forecast transformation, not a joint chance-constrained program. Marginal split-conformal coverage is expected under exchangeability; the chronological experiment reports empirical coverage rather than asserting that workplace sessions are exchangeable. Count under-prediction is measured but no fictitious session is synthesized because its physical window and energy would be unknown.

5.3 Receding-horizon information pattern

At interval t, the work set contains (i) actual connected sessions rebased to their remaining energy and (ii) forecasts whose predicted arrival is later than t. Arrived truth supersedes any overlapping forecast on the same charger. The controller solves the charger LP with the current monthly peak states, executes only the first action, updates delivered energy and (31), and repeats. Planned power is never delivered unless an actual EV is connected. There is no post-hoc replacement of missed energy. This follows the standard receding-horizon structure [9].

LPs can have many economically identical solutions. For reproducible EV/charger comparisons, every solve is lexicographic. First,

$$J^\star = \min_{x\in\mathcal{X}} J(x).$$

Then the controller chooses

$$x^\star_{\text{lex}} \in \arg\min_{x\in\mathcal{X}} \sum_{c,t} \frac{t}{T} p_{c,t} \quad \text{s.t.} \quad J(x) \le J^\star + \epsilon_{\text{lex}}, \tag{37}$$

where $\epsilon_{\text{lex}} = \max(10^{-6}, 10^{-7}|J^\star|)$ USD. The reported objective is always the primary tariff cost.

6 Experimental protocol

6.1 Data and leakage control

The local UCSD charging table contains 340,060 sessions from July 20, 2016 to September 16, 2024, covering 291 ports. The forecast benchmark uses 817 raw session records and 8,916.115 kWh on six fixed ports during July 1–September 30, 2023. The $h = 3$ matched cohort contains 372 training-eligible drivers, 239 test sessions, and 2,654.173 kWh. Thus it represents 29.25% of test sessions and 29.77% of test energy. Even $h = 1$ represents only 41.50% of test energy: most Q3 energy belongs to drivers with no training-period record at these ports.

The control pilot applies its physical and nonoverlap screens and uses 808 sessions over the same dates. Ports are ranked by activity and energy using May–June only; a data-quality screen additionally requires nonoverlap throughout the loaded evaluation window. This screen uses test-period timestamps only to verify the structural assumption and does not use test costs, energy magnitudes for ranking, or controller outcomes.

The flexibility-envelope experiment applies the physical screen consistently before training and evaluation. Across January 2022–September 2023 on the six ports, 120 infeasible-energy records and seven invalid/overnight windows are removed. Ninety-four overlapping port-days containing 474 records are excluded as structurally ambiguous, leaving 5,637 sessions. The resulting training-only $h = 3$ cohort contains 314 drivers. Its Q3 matched target is 2,642.384 kWh, or 29.75% of the physically screened full-demand total of 8,880.536 kWh. Q3 itself contains no overlapping port-days after the energy and window screen. The target identity in (17) is audited before every run; the maximum absolute discrepancy is 4.58×10^{-5} in float32 arithmetic.

Development and frozen-evaluation windows. The compact results reported in this draft are the already completed, chronologically frozen July–September evaluation. To avoid repeatedly spending three months of compute during model development, all new advanced forecast, flexibility, and coverage commands now default to July 1–31. Promotion of a future candidate to a paper claim requires an explicit frozen evaluation window and a new versioned result directory; July-only development outputs are never mixed into the existing Q3 tables.

Hyperparameter selection is chronological. Candidate $\alpha \in \{0.1, 0.2, 0.3, 0.4\}$ values use April–May residual calibration and June closed-loop validation. After selecting $\alpha = 0.1$, the corrections are refit on the immediately preceding May–June window, with the Q3 test untouched. The final 680 matched residuals give

$$q^a = 17 \text{ slots}, \qquad q^d = 18 \text{ slots}, \qquad q^E = 13.6035 \text{ kWh}.$$

Table 2: Concrete flexibility-task definitions. Individual EV and matched charger use the same 2,642.38 kWh of physically screened Q3 sessions, making entity resolution the only task-level difference. Full charger is the separate operational estimand.

Task	Forecast entity	Realized test scope	Entities	Q3 energy (kWh)	Coverage (%)
Individual EV	driver ID	recurring-driver sessions	314	2642.38	29.75
Matched charger	physical port	same recurring-driver sessions	6	2642.38	29.75
Full charger	physical port	all screened port sessions	6	8880.54	100.00

The 90th percentile count-underprediction residual is one session, but is reported rather than converted to a synthetic request.

6.2 Forecast methods and metrics

The forecasting comparison has three task labels. *EV* predicts each eligible driver separately. *ChargerMatched* aggregates exactly those drivers to the six ports. *ChargerFull* uses all sessions at the same ports. EV and ChargerMatched share the same realized energy by construction; ChargerFull measures operational station demand. We report per-entity count, energy, and timing errors for diagnostics, but the primary comparison uses aggregate daily energy MAE in (4). Full-scope deployability uses (6).

The hurdle model first classifies whether an entity is active and then regresses attributes conditional on activity. It is retained even when it fails because rare-event class weighting can generate widespread small positive predictions across hundreds of inactive drivers. GraphGNN is charger-only and is therefore reported for full-demand forecasting, not as a matched architecture pair.

The flexibility experiment uses the same chronological windows and 28-day lookback. Seasonal naive, unconstrained iTransformer followed by an explicitly reported projection diagnostic, and physics-iTransformer are evaluated. The physics model uses three seeds and six maximum epochs selected after a seven-day smoke test; the unconstrained diagnostic and deterministic seasonal baseline use one run. We report raw signature-validity rate and the mean absolute projection adjustment, so feasibility repair cannot be mistaken for native model accuracy. The physics model requires no inference-only repair.

6.3 Control methods and metrics

Six controllers share the same tariff, physical model, state update, and solver:

- **V0G**: immediate maximum-rate charging;
- **Perfect**: future actual sessions supplied as an unattainable information upper benchmark;
- **No forecast**: MPC knows a request only after it arrives;
- **Persistence**: sessions from the same weekday one week earlier;
- **Historical median**: the causal point forecast in (33); and
- **Conformal**: (36) applied to the historical median.

Cost savings and perfect-information regret are

$$S_m = 100\frac{J_m^{\mathrm{V0G}} - J_m}{J_m^{\mathrm{V0G}}}, \qquad R_m = 100\frac{J_m - J_m^{\mathrm{Perfect}}}{J_m^{\mathrm{Perfect}}}. \tag{38}$$

Forecast errors are computed on ordinal-matched sessions; count mean absolute error (MAE) is averaged per charger-day. A run is accepted only if unserved energy is below 10^{-5} kWh, the solver reports optimality at every call, fallback count is zero, and peak states are monotone within each month.

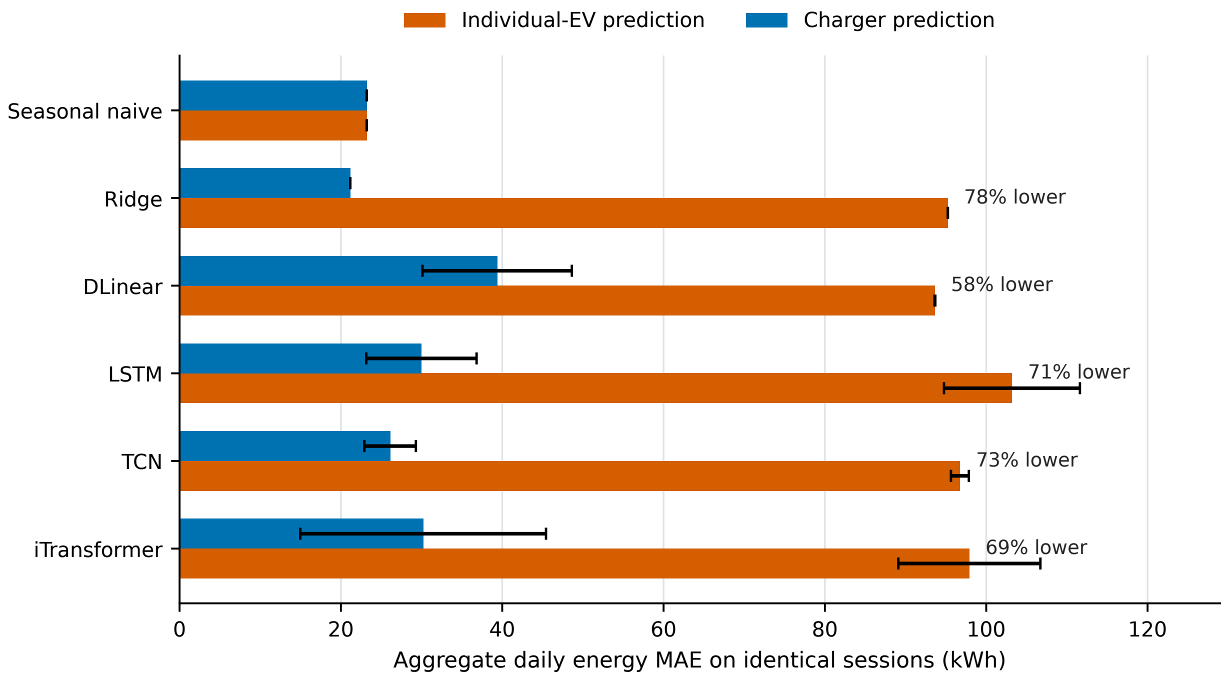

Figure 2: Aggregate next-day energy error on identical recurring-driver sessions. Bars are means over three seeds for neural models and one deterministic run for seasonal naive and ridge; error bars show seed standard deviation. Hurdle ridge is omitted from the plot to retain a readable scale but remains in Table 3.

7 Results

7.1 Fair EV-versus-charger forecasting

Table 3: Matched-scope Q3 forecasting. EV and charger columns predict identical realized sessions. "Reduction" is the charger MAE reduction relative to EV MAE. The final column is the 95% moving-block-bootstrap interval for paired daily EV-minus-charger absolute error.

Model	EV matched MAE (kWh/day)	Charger matched MAE (kWh/day)	Reduction (%)	95% block CI (kWh/day)
Seasonal naive	23.20	23.20	0.0	[-0.00, 0.00]
Ridge	95.24	21.17	77.8	[70.20, 79.57]
Hurdle ridge	1268.99	16.52	98.7	[1228.48, 1279.29]
DLinear	93.64	39.37	58.0	[51.42, 57.52]
LSTM	103.19	29.98	70.9	[68.79, 78.65]
TCN	96.74	26.10	73.0	[66.36, 76.07]
iTransformer	97.91	30.20	69.2	[61.87, 74.79]

Table 3 shows the consequence of entity resolution. Seasonal naive is equal to numerical precision at 23.20 kWh/day because its lag operator commutes with aggregation. This equality is a strong audit: neither side receives a different target or test-day filter. Ridge and all four neural architectures instead accumulate errors across 372 sparse driver series. Charger-level MAE is 57.95% lower for DLinear, 70.94% lower for LSTM, 73.02% lower for TCN, and 69.16% lower for iTransformer. Their paired block-bootstrap intervals for EV-minus-charger daily error are entirely positive. The intervals quantify Q3 day-level uncertainty conditional on this site, cohort, split, and model-fitting procedure; they do not establish cross-site generalization.

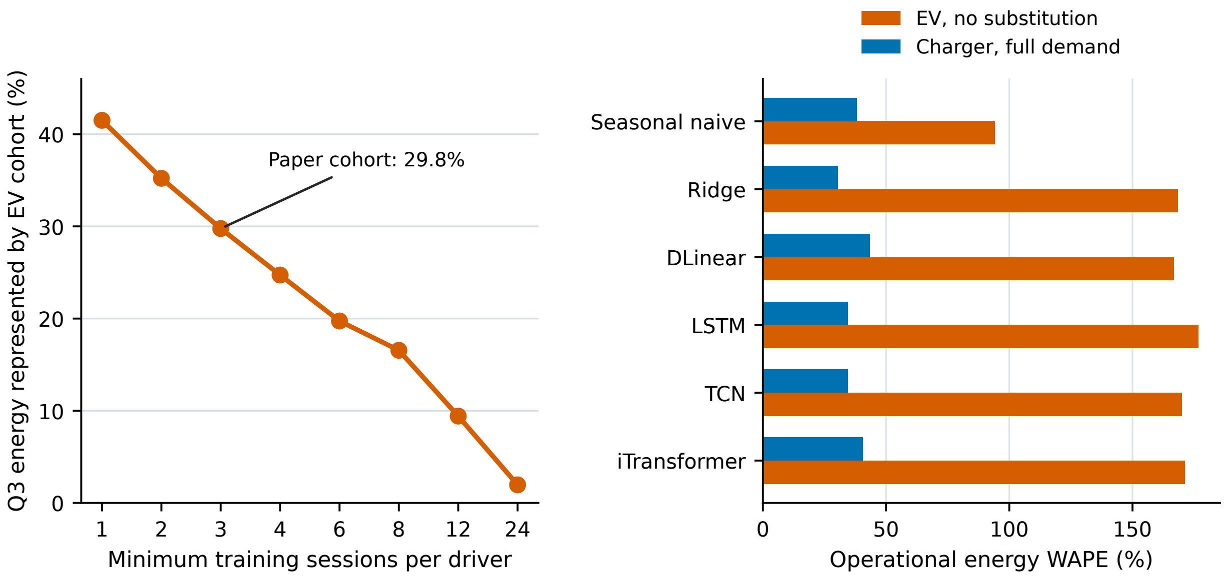

Figure 3: Operational scope. Left: Q3 energy represented as the minimum training-history requirement changes; entity selection uses training data only. Right: no-substitution EV WAPE versus full-demand charger WAPE. Overprediction assigned to one driver cannot cancel energy from an unidentified driver.

Table 4: Full six-charger Q3 next-day energy forecasting. Neural entries use three seeds; deterministic entries use one run.

Charger model	Seeds	Full-demand MAE (kWh/day)	Energy WAPE
Ridge	1	29.59	30.5%
GraphGNN	3	30.40	31.4%
LSTM	3	33.38	34.4%
TCN	3	33.45	34.5%
Seasonal naive	1	36.89	38.1%
Hurdle ridge	1	38.64	39.9%
iTransformer	3	39.23	40.5%
DLinear	3	42.04	43.4%

The hurdle result is a useful negative control. Balanced participation classification produces many small false-positive driver forecasts; after summation, EV MAE reaches 1,268.99 kWh/day even though per-driver errors look small. Aggregating first reduces this rare-event multiplicity and gives 16.52 kWh/day. This result is not evidence that all hurdle designs fail. Rather, it demonstrates why per-driver MAE and class-balanced accuracy are insufficient objectives for an aggregate control application.

Figure 3 separates prediction accuracy from deployability. The main $h = 3$ EV cohort misses 70.23% of full energy before any regression error is considered. Across the common models, its no-substitution operational WAPE ranges from 94.17% for seasonal naive to 176.70% for LSTM, whereas full-demand charger WAPE ranges from 30.53% to 43.37% for the corresponding non-hurdle baselines and neural models. The unreplicated ridge charger baseline has the lowest point estimate (29.59 kWh/day, 30.53% WAPE). Among three-seed models, GraphGNN is most stable at 30.40 ± 1.48 kWh/day and 31.37% WAPE.

This first advanced experiment is forecast-only. It establishes a statistical and operational motivation for charger-based prediction but does not yet establish economic value. Section 7.2 removes the mean-session reconstruction problem by forecasting an additive feasibility set, but

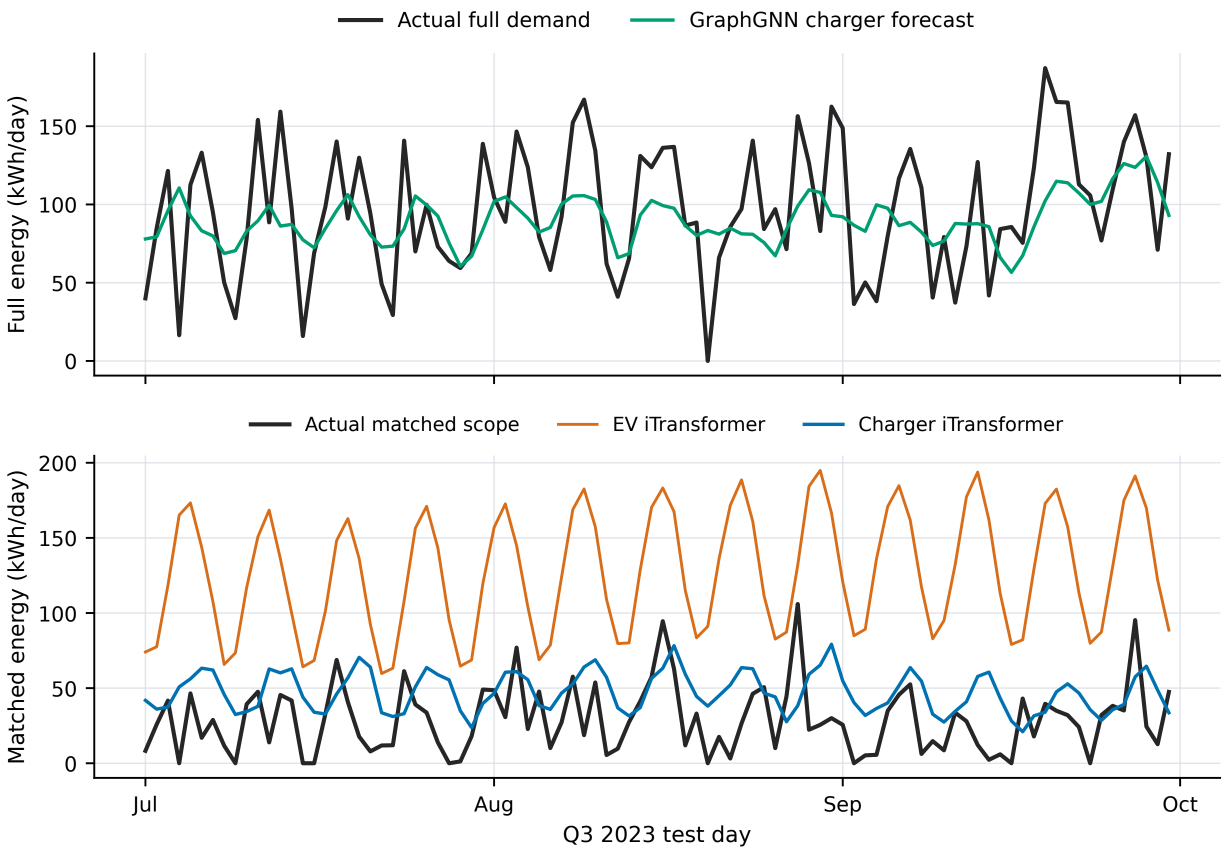

Figure 4: Q3 trajectories averaged over available seeds. Top: full-demand GraphGNN charger prediction. Bottom: iTransformer at both entity resolutions on the identical matched scope. Periodic positive bias across many sparse driver forecasts accumulates at EV resolution.

full-resolution uncertainty calibration and MPC coupling remain required before associating either error reduction with avoided cost.

7.2 Physics-constrained flexibility forecasting

Table 5 extends the matched identity from scalar energy to the feasible-energy set. Physics-iTransformer at charger resolution reduces terminal-energy, lower-envelope, and upper-envelope MAE by 69.19%, 60.71%, and 64.07%, respectively, relative to the same network trained on 314 sparse driver series. The paired 95% moving-block intervals are $[42.48, 53.07]$, $[18.56, 26.14]$, and $[23.12, 31.68]$ kWh, all strictly positive. This is not an artifact of a different realized target: seasonal-naive errors are identical at the two resolutions for every metric.

Unlike the earlier unconstrained architectures, the physics model produces a 100% valid raw-signature rate. Its matched charger errors are also lower than seasonal naive by 8.50% for terminal energy, 11.44% for the lower curve, and 12.27% for the upper curve. The result is important because the lower curve encodes deadline urgency and the upper curve encodes how much energy could already have been delivered; a terminal-energy-only model cannot distinguish these flexibility states.

Figure 6 shows that the EV error is persistent rather than caused by a small number of isolated days. The physics constraints therefore do not solve rare driver participation by themselves; they prevent impossible outputs, while charger aggregation addresses the sparse-identity estimation problem.

Table 6 separates three effects that a single accuracy number would hide. Seasonal naive is physically valid because it reuses an observed signature. The unconstrained iTransformer is com-

Table 5: Q3 matched-scope flexibility forecasting. Physics-iTransformer entries are mean ± seed standard deviation over three seeds. The bootstrap interval is for paired daily EV-minus-charger absolute error after seed averaging. Seasonal naive is identical at both resolutions.

Target	EV physics (kWh)	Charger physics (kWh)	Reduction (%)	95% block CI (EV minus charger, kWh)	Seasonal (kWh)
Terminal energy	68.74 ± 6.87	21.18 ± 2.54	69.2	[42.48, 53.07]	23.14
Lower envelope	36.73 ± 4.31	14.43 ± 0.98	60.7	[18.56, 26.14]	16.30
Upper envelope	42.62 ± 3.74	15.31 ± 0.98	64.1	[23.12, 31.68]	17.45

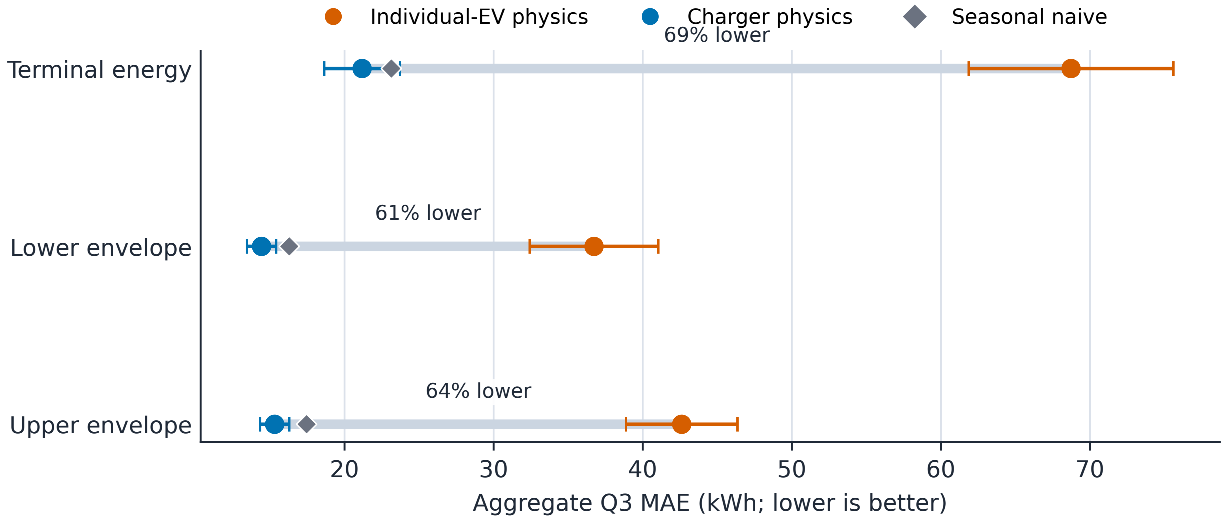

Figure 5: Matched-scope comparison as a connected-dot plot. Each line joins the same physics-iTransformer architecture at the two entity resolutions; horizontal bars are seed standard deviations. The diamond is the shared seasonal-naive error, and lower values are better.

petitive after projection but produces no valid raw signatures in these runs. Physics-iTransformer has the same learned backbone scale as its unconstrained counterpart and achieves 100% raw validity, so feasibility is an architectural property rather than a favorable repair step. The matched charger task also reduces the number of modeled entities from 314 to six without changing the realized target.

On full six-charger demand, physics-iTransformer gives 33.43 ± 4.84 kWh/day terminal-energy MAE versus 36.68 for seasonal naive. Its lower- and upper-curve MAE values are 24.57 ± 4.06 and 26.19 ± 3.13 kWh, respectively, versus 25.50 and 27.56 for seasonal naive. All three means improve, but individual seeds do not uniformly beat the deterministic baseline; the spread precludes claiming robust dominance. The one-seed unconstrained iTransformer has competitive errors only after projection; its raw valid-signature rate is zero, so it is not a deployable physical forecast without that repair.

The compressed signature is not yet a complete controller input. It must be lifted from six anchors to 96 slots, calibrated on validation residuals, and combined with arrived-session truth inside rolling MPC. Until that experiment is complete, Table 5 establishes a physical forecasting advantage, not a cost or revenue advantage.

7.3 Control value and forecast calibration

Table 8 shows that all MPC variants improve on V0G. The conformal controller reduces cost by 28.23% relative to V0G, closes most of the gap to perfect information, and costs 1.47% less than

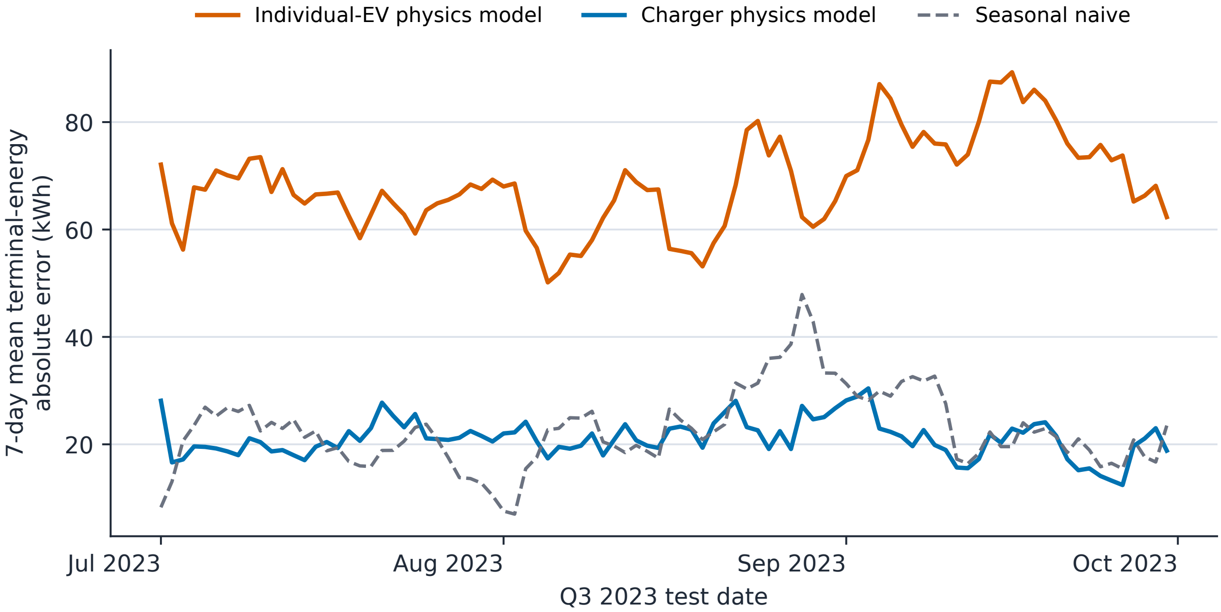

Figure 6: Seven-day moving mean of matched-scope terminal-energy absolute error through Q3 2023. Physics-model curves average three seeds. The same iTransformer is unstable across sparse driver identities, whereas charger error remains near the strong seasonal baseline.

Table 6: Model-level flexibility diagnostics across all three tasks. Values are aggregate MAE; stochastic entries are mean ± seed standard deviation. Best and second-best errors within each task are bold and underlined. "Raw valid" is evaluated before any projection.

Task	Model	Seeds	Terminal (kWh/day)	Lower (kWh)	Upper (kWh)	Width (kWh)	Raw valid (%)	Parameters
Individual EV	Seasonal naive	1	**23.14**	**16.30**	**17.45**	2.31	100	0
Individual EV	iTransformer + projection	1	76.32	45.26	50.64	6.03	0	14,331
Individual EV	Physics-iTransformer	3	<u>68.74 ± 6.87</u>	<u>36.73 ± 4.31</u>	<u>42.62 ± 3.74</u>	7.27	100	14,331
Physical charger	Seasonal naive	1	23.14	16.30	17.45	2.31	100	0
Physical charger	iTransformer + projection	1	<u>22.63</u>	<u>16.27</u>	<u>17.17</u>	2.44	0	14,331
Physical charger	Physics-iTransformer	3	**21.18 ± 2.54**	**14.43 ± 0.98**	**15.31 ± 0.98**	2.85	100	14,331
Full charger demand	Seasonal naive	1	36.68	25.50	27.56	5.44	100	0
Full charger demand	iTransformer + projection	1	<u>34.71</u>	**24.32**	**25.97**	6.00	0	14,331
Full charger demand	Physics-iTransformer	3	**33.43 ± 4.84**	<u>24.57 ± 4.06</u>	<u>26.19 ± 3.13</u>	5.43	100	14,331

historical-median MPC:

$$100\frac{4240.03 - 4177.90}{4240.03} = 1.465\%.$$

Its monthly cost improvement over historical median is \$50.32 in July and \$49.87 in August, but it is \$38.06 worse in September. Three monthly pairs are insufficient for a meaningful significance claim.

Historical median improves every point metric over persistence (Table 9). The robust transformation increases point MAE, which is expected because its purpose is one-sided risk protection rather than conditional-median accuracy. Against the historical-median point forecast, empirical one-sided coverage is 91.48% for late arrival, 91.48% for early departure, and 90.16% for energy underprediction, close to or above the 90% target.

The validation ranking in Table 10 selects the most conservative candidate. This does not imply that lower α must always be cheaper: shrinking windows and increasing energy can also create excessive early charging and higher peaks.

Table 7: Full six-charger flexibility forecasting. The unconstrained iTransformer is projected only for this diagnostic and has 0% raw valid signatures; physics-iTransformer is feasible by construction.

Full-demand model	Seeds	Terminal energy (kWh/day)	Lower envelope (kWh/anchor)	Upper envelope (kWh/anchor)
Seasonal naive	1	36.68	25.50	27.56
iTransformer + projection	1	34.71	24.32	25.97
Physics-iTransformer	3	33.43 ± 4.84	24.57 ± 4.06	26.19 ± 3.13

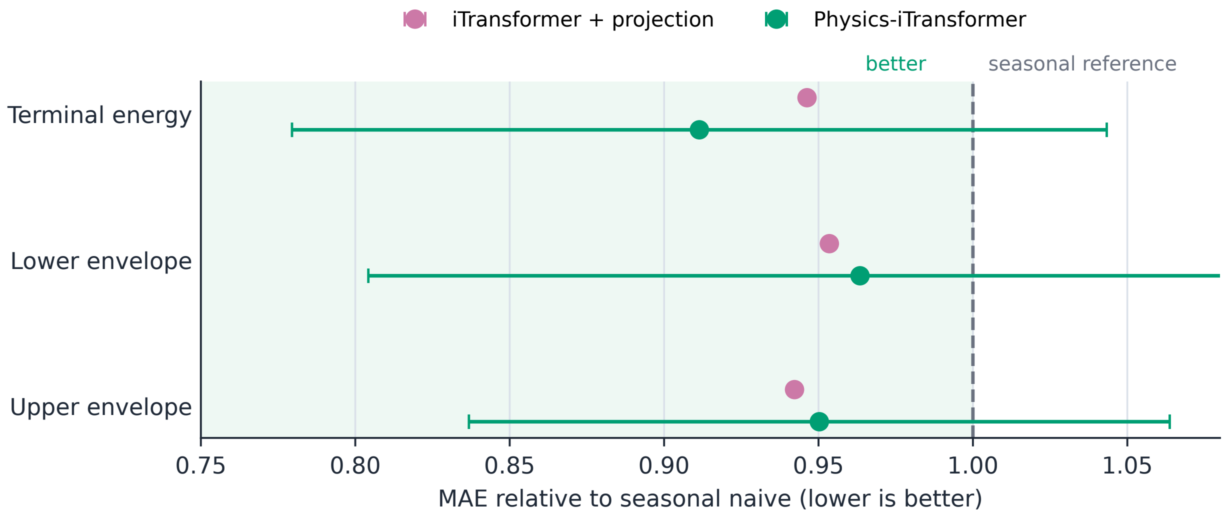

Figure 7: Full-demand Q3 error normalized by seasonal naive. Values left of the dashed reference improve on the baseline; horizontal bars show seed standard deviation after the same normalization. Physics-iTransformer improves all three means, but the intervals visibly cross the reference.

7.4 Exactness and computational scaling

On 28 real-day comparisons, EV- and charger-level LPs have maximum absolute objective difference 1.14×10^{-12} USD and maximum aggregate-load difference 7.64×10^{-11} kW after the common lexicographic tie-break. Every charger trajectory disaggregates to sessions with exact requested energy.

The charger formulation is not uniformly faster. It is slower at two sessions per port, approximately 1.18× faster at four, and 2.07× faster at eight. The relevant benefit is therefore density-dependent, not an unqualified aggregation speedup.

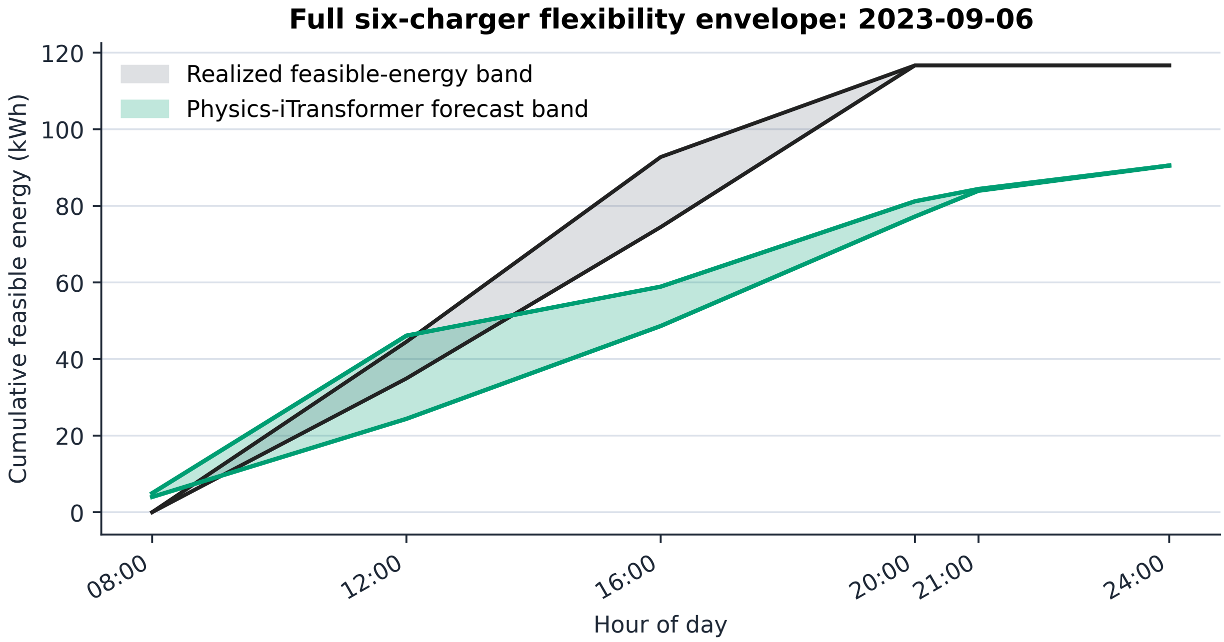

Figure 8: Representative full-demand day selected by median three-seed terminal error. Filled regions show realized and physics-iTransformer cumulative feasible-energy bands at the six anchors. Both predicted curves satisfy the physical cone despite the terminal-energy forecast error.

Table 8: Out-of-sample control results, July–September 2023. "Cost" is the sum of continuous monthly tariff contributions, not a sum of isolated daily demand bills.

Method	Q3 cost (USD)	Saving vs. V0G	Regret vs. perfect
V0G	5821.16	0.00%	45.50%
Perfect	4000.79	31.27%	0.00%
No forecast	4243.88	27.10%	6.08%
Persistence	4317.58	25.83%	7.92%
Hist. median	4240.03	27.16%	5.98%
Conformal	4177.90	28.23%	4.43%

Table 9: Q3 session forecast diagnostics. Arrival and departure MAE are in 15-minute slots. Robust inputs are intentionally not point-forecast-optimal.

Method	Count MAE (sessions)	Arrival MAE (slots)	Departure MAE (slots)	Energy MAE (kWh)
Persistence	0.962	10.783	11.744	8.031
Hist. median	0.868	9.366	10.687	6.811
Conformal	0.868	18.300	12.203	9.905

Table 10: Chronological June validation used to select α. June is never part of the Q3 test.

α	Nominal coverage	June validation cost (USD)
0.1	90%	1603.32
0.2	80%	1628.86
0.3	70%	1636.76
0.4	60%	1737.04

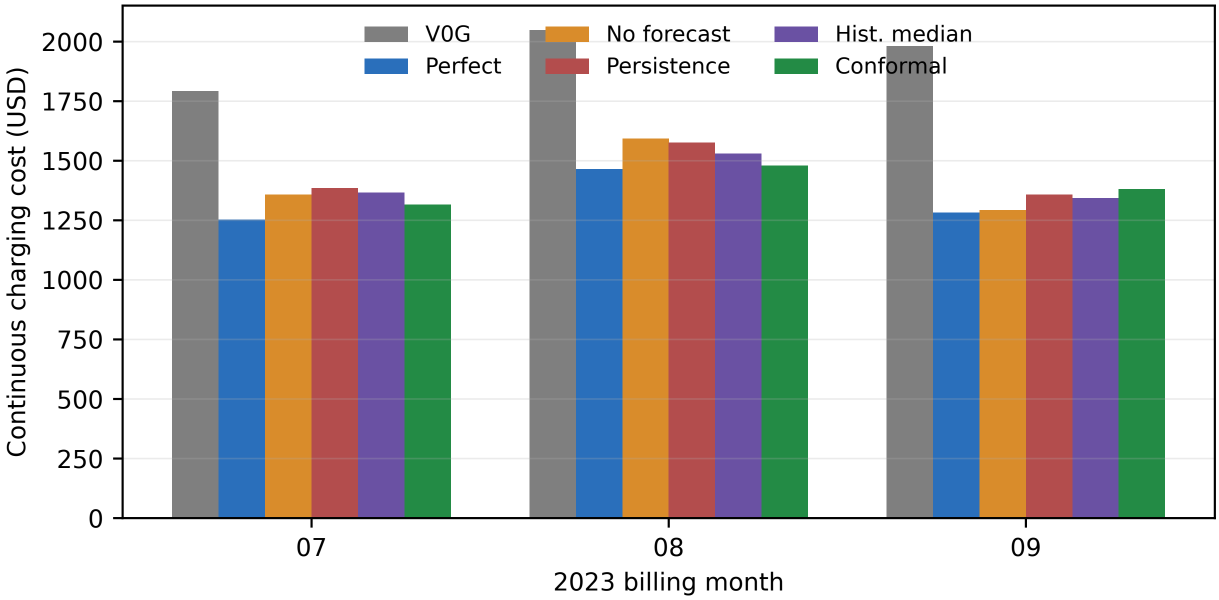

Figure 9: Continuous monthly charging cost. Each method's executed peak state is reset only at the billing-month boundary.

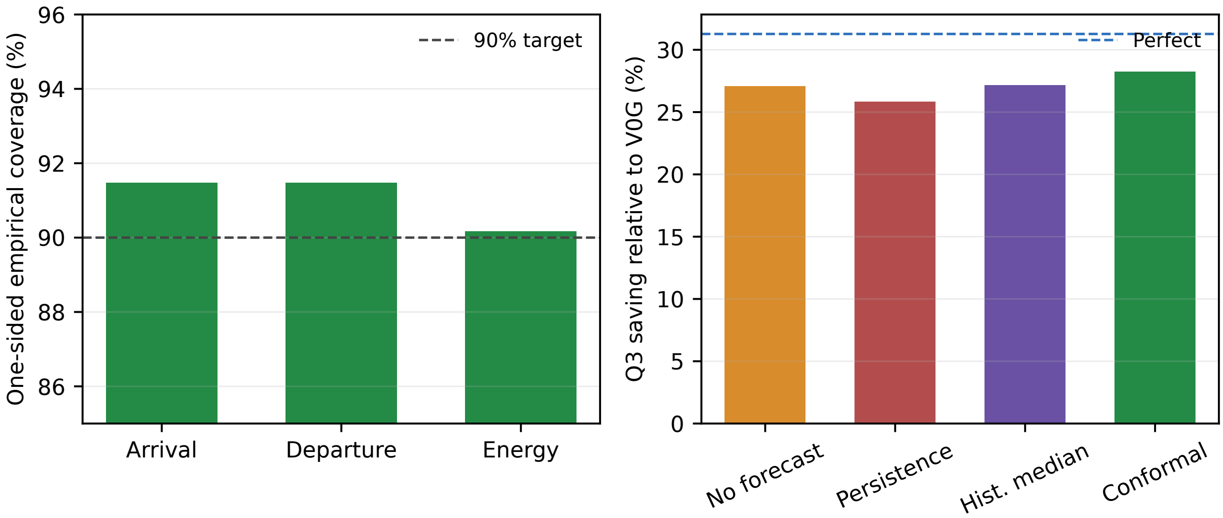

Figure 10: Left: empirical one-sided Q3 coverage of conformal corrections. Right: Q3 saving relative to V0G; the dashed line is perfect information.

Table 11: EV/charger invariance and synthetic scaling diagnostics.

Diagnostic	Value
Maximum absolute objective gap	1.14e-12
Maximum aggregate-load gap (kW)	7.64e-11
Median synthetic speedup at 8 sessions/charger	2.07×

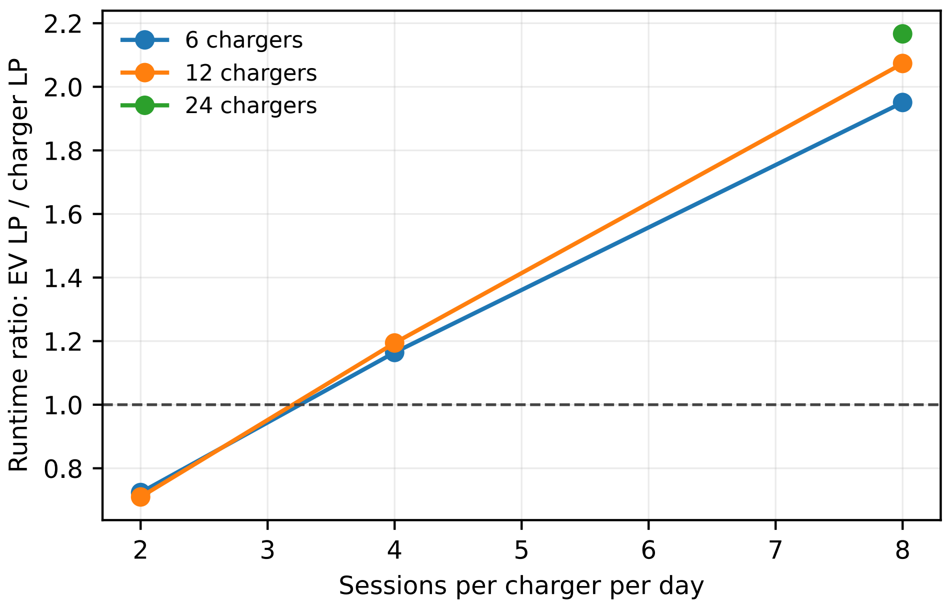

Figure 11: Median runtime ratio of the EV LP to the exact sparse charger LP. Aggregation becomes beneficial as sessions per charger increase; at low density, charger cumulative states can outweigh variable reduction.

8 Discussion and limitations

The experiments support six bounded conclusions. First, prediction entity is not a cosmetic modeling choice. On an identical session set, advanced driver-level errors accumulate across hundreds of sparse series, whereas the charger model learns the aggregate target directly. Second, a matched-scope metric alone is incomplete: open-set driver coverage limits the operational value of individual-EV forecasts when no reservation or mobility information is available. Third, billing state is part of the controller state: isolated-day demand accounting is not an innocuous reporting choice. Fourth, charger aggregation can be exact and computationally useful, but only because physical nonoverlap enables unambiguous disaggregation. Fifth, a simple calibrated uncertainty correction has positive Q3 economic value despite worse point MAE, illustrating that forecast quality should ultimately be evaluated through downstream control. Sixth, physical validity can be embedded in the forecast architecture: the flexibility head guarantees cumulative feasibility at every reported anchor, whereas all unconstrained neural signatures in the matched Q3 diagnostic require projection.

The seasonal-naive equality also clarifies what the results do not show. Aggregation alone cannot improve every forecasting rule: any linear forecast that commutes with summation and uses identical information can give the same aggregate output. The observed charger advantage arises when model fitting, regularization, nonlinear representation, or rare-event participation behaves differently at the two resolutions. This distinction is important for a theoretical revenue argument. The relevant quantity is not "charger versus EV" in isolation, but the interaction among entity resolution, estimator, uncertainty calibration, and the downstream control objective.

The evidence is deliberately bounded:

- six ports and three billing months do not establish statistical significance, seasonality robustness, or campus-wide generalization;
- the advanced forecast test uses one chronological Q3 split and three seeds rather than multiple years, sites, or rolling-origin folds;
- the individual-EV task has no reservation, employer schedule, mobility, or fleet-assignment data, so unseen-driver coverage is a limitation of the available information pattern rather than an impossibility theorem;
- correlation graph edges are learned from historical energy only; physical distance, electrical topology, traffic, and weather are not used;
- the new flexibility signatures are feasible at six anchors but have not been calibrated, lifted to 96 slots, or evaluated in rolling MPC, so their revenue effect is unknown;
- the cohort's nonoverlap requirement excludes shared-port anomalies and simultaneous connector use;
- calibration is marginal and ordinal-matched; it gives neither joint trajectory coverage nor protection against missing sessions;
- the model assumes energy and departure are known at plug-in and does not model early unplugging after arrival;
- tariff coefficients reproduce the research implementation and must be reconciled with the tariff vintage used in a final submission; and
- no V2G, demand response, network power flow, PV, or stationary storage is modeled. Accordingly, "revenue" claims would be misleading; the present quantity is avoided charging cost.

A submission-strength next experiment should pre-register multiple disjoint port cohorts, extend the untouched test to at least 12 billing months, and use rolling-origin folds. The predicted anchor cone should be lifted to full-resolution cumulative bounds using a validation-only monotone calibration map. It can then enter charger MPC directly as future aggregate flexibility; no forecast-to-realized EV assignment is required. Individual-EV and charger forecasts must enter their corresponding rolling MPCs under the identical tariff, information arrival, feasibility checks, and monthly peak state. Block bootstrap at the month or charger-month level can test both forecast and cost differences. That experiment can determine whether lower charger error creates stable avoided cost and whether the observed 1.47% conformal improvement is more than a Q3-specific pilot effect.

9 Conclusion

We separated individual-EV and physical-charger prediction into two auditable tasks and compared them on exactly the same sessions. Across common advanced architectures, charger resolution substantially reduces aggregate daily energy error, while the full-demand analysis exposes the additional open-set coverage burden of driver-specific prediction. We then replaced mean-session output with a set-free cumulative flexibility target and embedded its physical cone in iTransformer. On matched Q3 data, the charger physics model reduces all three envelope errors relative to both the same EV model and seasonal naive, while retaining 100% valid signatures. We also developed a forecast-calibrated charger MPC that preserves individual-session feasibility and carries monthly demand peaks correctly. In the six-port control pilot, conformal robustness reduces cost relative to immediate charging and modestly improves on the causal point forecast with zero unserved energy. Exact EV/charger optimization invariance and density-dependent scaling are independently verified. The combined evidence supports charger resolution as a promising prediction-and-control abstraction, but the causal link from the new physical forecast to economic value remains preliminary until the anchor cone is calibrated, lifted to full resolution, and run through MPC across more cohorts and seasons.

Reproducibility statement

Source code, compact result tables, experiment commands, generated figures, and this manuscript are version controlled. The raw and processed charging tables are excluded from Git because of size and data-use restrictions. Forecast cohort selection, scaling, graph construction, and early stopping are chronological. The MPC runner uses a fixed port cohort, chronological calibration/validation/test windows, daily checkpoints, and hard failures for unserved energy, solver fallback, or invalid physical overlap.

Data availability

The charging records are held locally and their redistribution is subject to UC San Diego and source-data restrictions. Derived compact numerical results needed to reproduce all manuscript tables and figures are included with the code. Access to the underlying session table should be requested through the appropriate data owner.

Declaration of competing interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported here.

Declaration of generative AI and AI-assisted technologies

During preparation of this draft, the author used an AI coding and writing assistant to help refactor code, generate tests, and edit language. All model formulations, numerical results, citations, and conclusions remain subject to the author's review and responsibility. This statement should be updated to match the journal's policy and the final author workflow before submission.

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A Implementation-to-equation audit

Traceability between the mathematical model and implementation.

Component	Equations	Implementation invariant
Matched forecasting	(1)–(4)	training-only driver cohort and exact daily energy identity
Operational scope	(5)–(6)	known-driver error and unidentified demand cannot cancel
Advanced models	(7)–(13)	training-only scaling/adjacency, fixed split, deterministic seeds
Flexibility forecast	(14)–(24)	same-session signature identity, differentiable physical output, raw validity audit, and no session assignment metric
Session feasibility	(26)	validation, exact energy equality, and availability bounds
Charger projection	(28)–(30)	nonoverlap assertion, sparse cumulative state, and explicit disaggregation
Monthly billing	(31)–(32)	executed peak carried day to day and reset only at month boundary
Conformal correction	(34)–(36)	finite-sample order statistic and capacity clipping
Rolling information	Section 5.3	arrived truth replaces future forecast; only connected actual load is executed
Tie-breaking	(37)	primary HiGHS LP followed by a constrained secondary LP

B Artifact commands

```
MPLCONFIGDIR=/private/tmp/mpl-models PYTHONPATH=src \
  .venv/bin/python -m unittest discover -s tests -v
PYTHONPATH=src .venv/bin/python \
  experiments/forecast_coverage_sensitivity.py
MPLBACKEND=Agg MPLCONFIGDIR=/private/tmp/mpl-fair PYTHONPATH=src \
  .venv/bin/python experiments/make_fair_forecast_artifacts.py
MPLCONFIGDIR=/private/tmp/mpl-envelope PYTHONPATH=src \
  .venv/bin/python experiments/envelope_forecast_benchmark.py --quick
MPLBACKEND=Agg MPLCONFIGDIR=/private/tmp/mpl-flex PYTHONPATH=src \
  .venv/bin/python experiments/make_flexibility_artifacts.py
bash experiments/run_paper_protocol.sh
```